2026 VCE Specialist Mathematics 2 (NHT) external assessment report

Section A – Multiple-choice questions

Question	Correct answer	Comments
1	D	Contrapositive statement required.
2	B	$f(x) = p$ $\lim_{x \to 4} f(x) = \frac{\pi}{80}$ $p = \frac{\pi}{80}$
3	C	$left$ $area$ 0 $2\pi f\left(\frac{0+1}{2}\right)\sqrt{(f(0)-f(1))^2+1} = 7.603$ 1 $2\pi f\left(\frac{1+2}{2}\right)\sqrt{(f(1)-f(2))^2+1} = 14.662$ $print = 7.603 + 14.662$
4	D	$x^3 - x^2 - m^2x + m^2 = (x-1)(x^2-m^2)$ 3 vertical asymptotes
5	B	The maximum value of $\lvert z \rvert$ is the furthest point that lies on $\lvert z-2-i \rvert = 3$ from origin. A ray from the origin to this point must pass through the centre of the circle. Distance = radius + distance from origin to the circle centre.
6	D	Take care with inequalities. Dotted lines indicate excluded values; solid lines indicate that values are included. The required region lies between the arguments of $\frac{\pi}{3}$ and $\frac{\pi}{2}$ and between the circles with radius 1 and radius 2.
7	B	Exclude C and D as there are not vertical tangents on the line $y = x$. The sign of gradient is positive when $x > y$.

Question	Correct answer	Comments
8	C	$\frac{dy}{dx}=\frac{dy}{dt}\div\frac{dx}{dt}$ Be careful to use exact values.
9	A	$x^3=u^2+1\Rightarrow 3x^2=2u\frac{du}{dx}$ Change terminals to match the u-substitution.
10	B	$area=2\pi\int_0^{\pi}(1-\cos(2t))\sqrt{(2-2\cos(2t))^2+(2\sin(2t))^2}\,dt$ Simplify and use double angle formulas.
11	A	$\frac{dy}{dx}=\frac{e^x e^y\sin(2x)}{\sin(x)\cos(y)-\cos(x)\sin(y)-\sin(x)\cos(y)}$ $\frac{dy}{dx}=\frac{e^x\sin(2x)}{-\cos(x)}\times\frac{e^y}{\sin(y)}$ Double angle formula
12	D	$\frac{dt}{dP}=\frac{50\,000}{P(1000-P)}$ Initial population is 50, want time until $P=500$.
13	C	Changes direction when velocity = 0. $v(t)=(\cos(t)-\sin(t))e^{-t}$ $v=0\ when\ t=\frac{\pi}{4},\ \frac{5\pi}{4},\ \frac{9\pi}{4}$
14	D	$a=v\frac{dv}{dx}$ $x=\int_4^6\frac{v}{2+v}\,dv$
15	C	$\underset{\sim}{u}\cdot\hat{\underset{\sim}{v}}=2$ $\frac{2a+6-5}{3}=2$
16	B	$\underset{\sim}{a}\cdot\left(\underset{\sim}{b}-\underset{\sim}{c}\right)=\sqrt{p^2+17}\sqrt{42}\cos(\theta)$ $p-4+20=\sqrt{p^2+17}\sqrt{42}\times\frac{1}{2}$

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17	B	$\underset{\sim}{r}(t)=(t+1)\underset{\sim}{i}+(\sin(t))\underset{\sim}{j}+(1-\cos(t))\underset{\sim}{k}$ $\underset{\sim}{s}(t)=t\underset{\sim}{i}+(\sin(2t))\underset{\sim}{j}+(\cos(t)-2)\underset{\sim}{k}$ $\left\lvert\underset{\sim}{r}(2\pi)-\underset{\sim}{s}(2\pi)\right\rvert$
18	A	Find angle between two normals and subtract from π .
19	A	Solve equations simultaneously to find parametric solution.
20	C	$\mathrm{E}(Q)=15c+50$ $\mathrm{Var}(Q)=16c^2+36$ Use inverse normal distribution to obtain $z=1.281\ldots$ $\frac{112.2-(15c+50)}{\sqrt{16c^2+36}}=1.281\ldots$

Section B

Question 1a.

$$-\frac{1}{2}x+1+\frac{2x}{4-x^2}$$

Question 1b.

All three asymptotes were required.

$$y=-\frac{1}{2}x+1,\quad x=-2,\quad x=2$$

Question 1c.

A CAS question for 1 mark.

$$f'(x)=\frac{x^2(12-x^2)}{2(4-x^2)^2}\quad\text{OR}\quad\frac{-x^2\left(x^2-12\right)}{2\left(x^2-4\right)^2}\quad\text{OR}\quad\frac{2\left(x^2+4\right)}{\left(x^2-4\right)^2}-\frac{1}{2}\quad\text{OR}\quad-\frac{1}{2}+\frac{1}{(x+2)^2}+\frac{1}{(x-2)^2}$$

Question 1di.

$$\left(-2\sqrt{3}, \frac{2+3\sqrt{3}}{2}\right), \left(2\sqrt{3}, \frac{2-3\sqrt{3}}{2}\right)$$

Many students incorrectly included the coordinates for the stationary point of inflection.

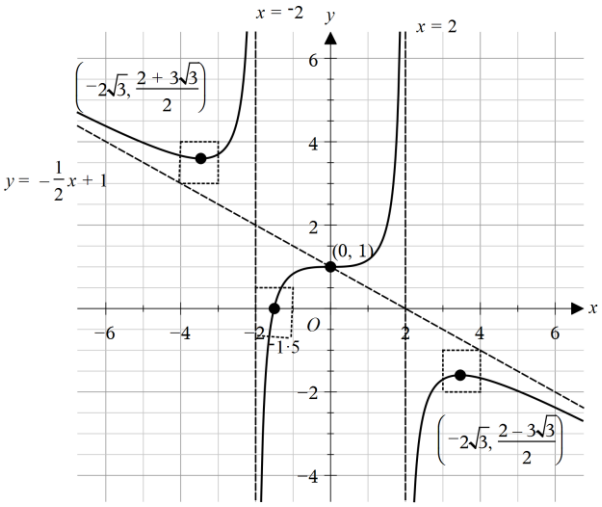
Question 1dii.

$(0, 1)$

Question 1e.

Care should be taken with:

- the position of labelled points – they should be within the box shown in the sketch below
- the y-intercept as it must have zero gradient
- asymptotic behaviour.



Question 1f.

$k < 0$

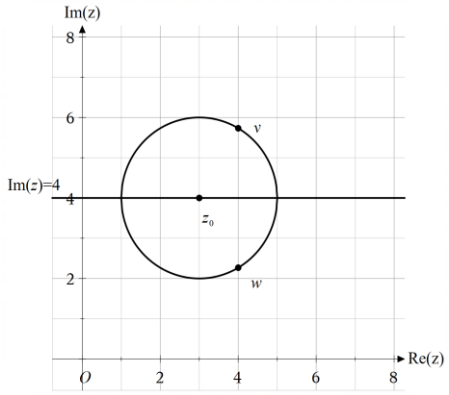
Question 1gi.

$$y = -\frac{1}{2}x - \frac{b}{2}$$

Question 1gii.

$b = 0, -4$

Question 2a.



Many did not draw and label the line $\text{Im}(z) = 4$.

Question 2b.

$$(x-3)^2+(y-4)^2=2^2$$

Substitute $x=4$, $1+(y-4)^2=4$

Solves to give $y=4\pm\sqrt{3}$, so $\text{Im}(v)=4+\sqrt{3}$

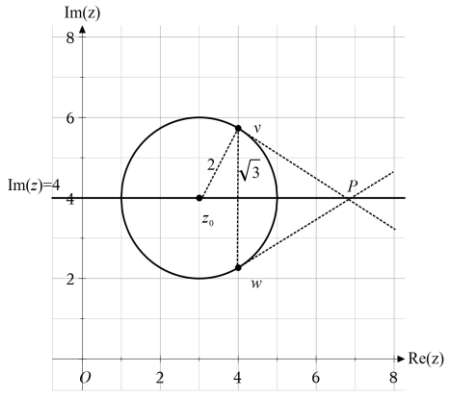
Question 2c.

Angle subtended at z_o by arc is $\dfrac{4\pi}{3}$

$$A=\frac{1}{2}\times 2^2\times\left(\frac{4\pi}{3}-\sin\frac{4\pi}{3}\right)$$

$$A=\frac{8\pi}{3}+\sqrt{3}$$

Question 2d.



$\frac{2}{d} = \cos\frac{\pi}{3},\ d = 4$

OR use tangent equations: $y = -\frac{1}{\sqrt{3}}x + 4 + \sqrt{3} + \frac{4}{\sqrt{3}},\quad y = \frac{1}{\sqrt{3}}x + 4 - \sqrt{3} - \frac{4}{\sqrt{3}}$

OR solve either tangent intersecting $y = 4$

x-coordinate of P is 7.

Question 2e.

$|z-(6+4i)|=1$

$|z-(4+4i)|=3$

Question 3ai.

$\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt},\quad \frac{dy}{dx} = y+1$

Question 3aii.

Sufficient working must be shown for a 'show that' question.

$x = \int \frac{1}{y+1}\,dy,\quad x = \log_e(y+1) + c,\quad y(0) = 0, \Rightarrow c = \log_e 1,\ c = 0$

Or might have

$\int_0^y \frac{1}{y+1}\,dy = \int_0^x 1\,dx$

$\log_e(y+1) = x,\ y+1 = e^x$ so $y = e^x - 1$.

Question 3bi.

'Use calculus' indicates that the antiderivative must be shown in the solution.

$$\pi\int_0^{\log_e k}\left(e^x-1\right)^2 dx=\pi\log_e k$$

$$\pi\int_0^{\log_e k}\left(e^{2x}-2e^x+1\right)dx=\pi\log_e k$$

$$\left[\frac{1}{2}e^{2x}-2e^x+x\right]_0^{\log_e k}=\log_e k$$

Integrate and equate to the given volume.

The antiderivative needed to be included in the square brackets.

$$\frac{1}{2}k^2-2k+\log_e k-\frac{1}{2}+2=\log_e k$$

$$k^2-4k+3=0$$

Question 3bii.

$k^2-4k+3=0$ solves to give $k=3\ \left(\text{as } k>1\right)$

Question 3ci.

$$x=t,\quad y=e^t-1,\quad y+1=e^t$$

$$L=\int_0^{\log_e(5)}\sqrt{1^2+\left(e^t\right)^2}\,dt$$

$$L=\int_0^{\log_e(5)}\sqrt{1^2+e^{2t}}\,dt$$

Question 3cii.

$$\int_0^{\log_e(5)}\sqrt{1+e^{2t}}\,dt=4.4$$

Question 3di.

$$A = 2\pi\int_0^{\log_e(5)} t \times \sqrt{1^2 + \left(e^t\right)^2}\, dt$$

Or in Cartesians:

$$A = 2\pi\int_0^4 \log_e\left(y+1\right) \times \sqrt{1 + \left(\frac{1}{y+1}\right)^2}\, dy \quad \text{OR} \quad A = 2\pi\int_0^{\log_e(5)} x \times \sqrt{1^2 + \left(e^x\right)^2}\, dx$$

Question 3dii.

$A = 26.9$

Question 4ai.

$$\underset{\sim}{V}_{D_1} = 2\underset{\sim}{i} + 3\underset{\sim}{j} + 5\underset{\sim}{k}, \quad \left|\underset{\sim}{V}_{D_1}\right| = \sqrt{38} \simeq 6.2$$

Question 4aii.

$$\underset{\sim}{D}_1(2) = 5\underset{\sim}{i} + 7\underset{\sim}{j} + 10\underset{\sim}{k}, \ \underset{\sim}{D}_2(2) = 10\underset{\sim}{i} + 22\underset{\sim}{j} + 16\underset{\sim}{k}$$

$$50 + 154 + 160 = \sqrt{174} \times \sqrt{840} \times \cos\theta$$

Scalar product applied.

$$\theta = 17.8^\circ$$

Question 4b.

$$\underset{\sim}{D}_2(t) - \underset{\sim}{D}_1(t) = \left(3t-1\right)\underset{\sim}{i} + \left(7t+1\right)\underset{\sim}{j} + \left(10-2t\right)\underset{\sim}{k}$$

$$\left|\underset{\sim}{D}_2(t) - \underset{\sim}{D}_1(t)\right|^2 = \left(3t-1\right)^2 + \left(7t+1\right)^2 + \left(10-2t\right)^2 \left(= 62t^2 - 32t + 102\right)$$

Minimise to give $t = \frac{8}{31} \simeq 0.258$, and Distance = 9.893

Question 4ci.

$$\underset{\sim}{D}_3(t) \times \underset{\sim}{D}_4(t) = \left(1-t\right)\underset{\sim}{i} + \left(t-1\right)\underset{\sim}{k}$$

Question 4cii.

$\underset{\sim}{D}_3(1)\times\underset{\sim}{D}_4(1)=\underset{\sim}{0}$ null vector – requires tilde or equivalent vector notation.

Means position vectors lie on the same line.

Line from the origin to one drone is the same line from the origin to the other drone.

Question 4d.

$\underset{\sim}{V}_{D_3}=\underset{\sim}{i}+\underset{\sim}{k}, \quad \left|\underset{\sim}{V}_{D_3}\right|=\sqrt{2}$

$\underset{\sim}{V}_{d_4}=\underset{\sim}{i}+(\cos t-\sin t)\underset{\sim}{j}+\underset{\sim}{k}, \quad \left|\underset{\sim}{V}_{d_4}\right|=\sqrt{2+(\cos t-\sin t)^2}$

$$\sqrt{2}=\sqrt{2+(\cos t-\sin t)^2}$$

Solves to give $t=\frac{\pi}{4}+k\pi$ where $k\in Z^+\cup\{0\}$. Accepted N or $N\cup\{0\}$

Question 5a.

$\overrightarrow{P_1P_2}=-\frac{4}{3}\underset{\sim}{j}+\frac{4}{3}\underset{\sim}{k}, \ \overrightarrow{P_1P_3}=-\frac{4}{3}\underset{\sim}{i}+\frac{2}{3}\underset{\sim}{j}+2\underset{\sim}{k}$ Find two spanning vectors

$\overrightarrow{P_1P_2}\times\overrightarrow{P_1P_3}=-\frac{32}{9}\underset{\sim}{i}-\frac{16}{9}\underset{\sim}{j}-\frac{16}{9}\underset{\sim}{k}$ Apply a cross product using spanning vectors

Area $=\frac{1}{2}\times\left|\overrightarrow{P_1P_2}\times\overrightarrow{P_1P_3}\right|=\frac{8\sqrt{6}}{9}$

Question 5b.

$2x+y+z=4$ For their left side using their normal from above.

Question 5ci.

$$\frac{1}{2}\times\left|\overrightarrow{P_1P_2}\times\overrightarrow{P_1P_3}\right|=\frac{8\sqrt{6}}{9}$$

Question 5cii.

$$\left|\left(6-\frac{2d}{3}\right)\underset{\sim}{\mathrm{k}}.\left(\frac{2\underset{\sim}{\mathrm{i}}-\underset{\sim}{\mathrm{j}}+\underset{\sim}{\mathrm{k}}}{\sqrt{6}}\right)\right|=2\sqrt{6}$$

Solves to give $d=-9$ and $d=27$

Question 5d.

$\left(2\underset{\sim}{\mathrm{i}}-\underset{\sim}{\mathrm{j}}+\underset{\sim}{\mathrm{k}}\right).\left(p^2\underset{\sim}{\mathrm{i}}-(\mathrm{p}+2)\underset{\sim}{\mathrm{j}}-5\underset{\sim}{\mathrm{k}}\right)=0$ Scalar product of normals equated to 0.

$2p^2+p+2-5=0,\ 2p^2+p-3=0$

Solves to give $p=-\frac{3}{2}$ and $p=1$.

Question 6a.

$T_1-T_2\sim \mathrm{N}\left(0,\ 2\times 5^2\right)$

$\Pr\left(-5<T_1-T_2<5\right)=0.5205$

Question 6b.

$\overline{T}\sim \mathrm{N}\left(200,\ \frac{25}{4}\right)$

$\Pr\left(195<\overline{T}<205\right)=0.9545$

Question 6c.

$H_0:\ \mu=200$, $H_1:\ \mu<200$

Question 6di.

$\overline{T}\sim \mathrm{N}\left(200,\ \frac{25}{25}\right),\quad \Pr\left(\overline{T}<198.5\,|\,\mu=200\right)=0.0668$

Question 6dii.

No improvement as p value is greater than 0.05.

Question 6e.

$\Pr\left(\overline{T} < \bar{t}_{\text{Critical}} \mid \mu = 200\right) = 0.05$

$\Rightarrow \bar{t}_{\text{Critical}} = 198.3551$

Need $\Pr\left(\overline{T} > \bar{t}_{\text{Critical}} \mid \mu = 196\right)$

$= 0.01$